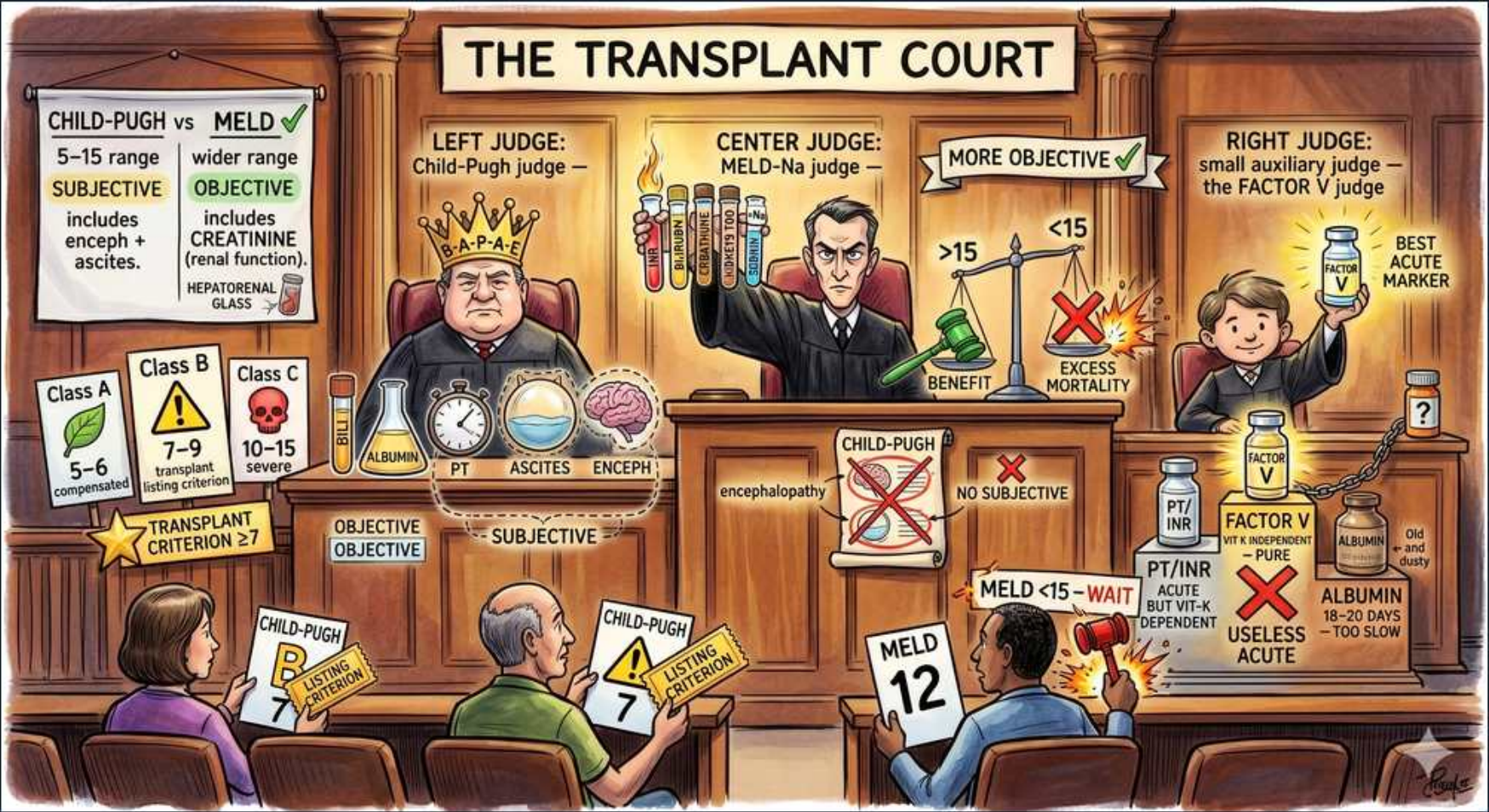


Liver Approach — Transplant Court



1. Child-Pugh vs MELD board

WHAT YOU SEE

Top-left placard 'CHILD-PUGH vs MELD', 5-15 range vs wider range

WHAT IT ENCODES

The two cirrhosis severity scores are Child-Pugh (CTP) and MELD. Child-Pugh ranges 5-15 points and contains SUBJECTIVE clinical items (ascites, encephalopathy graded by the clinician). MELD is fully OBJECTIVE (laboratory-only) and ranges ~6-40, which is why it (as MELD-Na) is used for organ ALLOCATION/waitlist prioritization, while Child-Pugh is a bedside prognostic estimate.

BOARD TRAP

WRONG answer: using Child-Pugh to prioritize the transplant waiting list. Discriminator: organ allocation uses the objective MELD-Na (creatinine, bilirubin, INR, sodium); Child-Pugh's subjective ascites/encephalopathy make it unsuitable for ranking candidates fairly.

2. Child-Pugh subjective items

WHAT YOU SEE

'includes enceph + ascites' note

WHAT IT ENCODES

Child-Pugh's two SUBJECTIVE components are ascites and hepatic encephalopathy, each graded none/mild/moderate-severe. These clinician-dependent items are precisely what MELD deliberately leaves out to keep allocation objective.

BOARD TRAP

WRONG answer: thinking MELD includes ascites or encephalopathy. Discriminator: ascites and encephalopathy belong to CHILD-PUGH only; MELD uses creatinine, bilirubin, INR (+Na for MELD-Na) — no clinical/subjective variables.

3. MELD objective items

WHAT YOU SEE

'includes creatinine (renal), bilirubin, INR'

WHAT IT ENCODES

MELD is calculated from three objective labs — creatinine (renal function), total bilirubin, and INR — and MELD-Na adds serum sodium because hyponatremia independently predicts mortality in cirrhosis. The inclusion of creatinine is why hepatorenal syndrome sharply raises the MELD score.

BOARD TRAP

WRONG answer: omitting sodium or adding albumin to MELD. Discriminator: modern allocation uses MELD-Na (Cr, bilirubin, INR, Na); albumin is a Child-Pugh component, NOT a MELD variable.

4. Child classes A/B/C

WHAT YOU SEE

'Class A 5-6, Class B 7-9, Class C 10-15'

WHAT IT ENCODES

Child-Pugh classes: A = 5-6 points (well-compensated, ~1-year survival ~100%), B = 7-9 (significant compromise, ~80%), C = 10-15 (decompensated, ~45%). The class guides surgical risk and broad prognosis.

BOARD TRAP

WRONG answer: assuming a Child-Pugh A patient is safe for major elective abdominal/hepatic surgery without further thought. Discriminator: even Child A carries elevated perioperative mortality, and Child C is often a contraindication to elective surgery — use the class to assess operative risk, and MELD for waitlist priority.

5. Transplant criterion ≥7

WHAT YOU SEE

'TRANSPLANT CRITERION ≥7'

WHAT IT ENCODES

A Child-Pugh score ≥7 (class B) is the traditional threshold to begin transplant EVALUATION/referral, marking the transition from compensated to clinically significant disease. It flags who should be referred, not who is immediately listed.

BOARD TRAP

WRONG answer: treating Child-Pugh ≥7 as the score that determines actual list priority/timing. Discriminator: ≥7 triggers referral for evaluation, but listing and ranking are by MELD-Na — once referred, the objective MELD score (esp. ≥15) drives timing.

6. Left judge: Child-Pugh

WHAT YOU SEE

Crowned 'LEFT JUDGE: Child-Pugh judge' with PT/albumin/ascites/enceph

WHAT IT ENCODES

The Child-Pugh 'judge' combines five components: prothrombin time/INR, albumin, bilirubin, ascites, and encephalopathy. Two are objective labs (PT, albumin, bilirubin) and two are subjective (ascites, encephalopathy) — the blend is why it is a prognostic, not an allocation, tool.

BOARD TRAP

WRONG answer: leaving out bilirubin or albumin when listing Child-Pugh components. Discriminator: remember all five via PABASE (PT, Albumin, Bilirubin, Ascites, Encephalopathy); dropping bilirubin/albumin is the common partial-recall error.

7. PABASE mnemonic

WHAT YOU SEE

Crown labeled PABASE / hepatocellular glass

WHAT IT ENCODES

PABASE = PT (INR), Albumin, Bilirubin, Ascites, Encephalopathy — the five Child-Pugh components. Note bilirubin and INR are SHARED with MELD, while albumin/ascites/encephalopathy are unique to Child-Pugh.

BOARD TRAP

WRONG answer: adding creatinine or sodium to PABASE. Discriminator: creatinine and sodium are MELD-Na variables, NOT Child-Pugh — Child-Pugh has no renal term, which is one reason MELD (with creatinine) better captures the sickest, often hepatorenal, patients.

8. Center judge: MELD-Na

WHAT YOU SEE

'CENTER JUDGE: MELD-Na judge', tubes of creat/bili/INR

WHAT IT ENCODES

The MELD-Na 'judge' prioritizes deceased-donor organ ALLOCATION using objective labs (creatinine, bilirubin, INR, sodium). Because it is purely quantitative it is harder to game and ranks candidates by predicted short-term mortality — the sickest objective scores go first.

BOARD TRAP

WRONG answer: thinking the patient with the worst symptoms (massive ascites, encephalopathy) automatically gets priority. Discriminator: allocation is by MELD-Na numbers, not symptom burden; some severely symptomatic Child-Pugh patients can still have a relatively low MELD, and 'exception points' (e.g., for HCC) are added separately.

9. MELD predicts mortality

WHAT YOU SEE

Scales: '>15 benefit', '<15 excess mortality'

WHAT IT ENCODES

MELD predicts 90-day mortality on the waitlist and is the basis for the transplant-benefit threshold: survival benefit from transplantation generally emerges when MELD >15, whereas transplanting at MELD <15 tends to incur EXCESS mortality compared with waiting (the operative/immunosuppression risk outweighs the disease risk).

BOARD TRAP

WRONG answer: listing/transplanting a low-MELD (<15) patient on the assumption that earlier transplant is always better. Discriminator: below MELD 15 the surgical/immunosuppression risk exceeds the benefit — these patients do better waiting; benefit accrues above 15.

10. MELD <15 = WAIT

WHAT YOU SEE

'MELD <15 - WAIT' card, gavel

WHAT IT ENCODES

A MELD <15 means outcomes are better by continuing to WAIT and monitor rather than transplanting now — the predicted post-transplant risk outweighs the benefit. These patients are managed medically with ongoing surveillance and re-scoring.

BOARD TRAP

WRONG answer: placing a MELD <15 patient on the active transplant list based on the score alone. Discriminator: a low MELD argues for waiting; the exception is HCC or other 'MELD exception' indications, where standardized exception points override the raw lab MELD.

11. MELD 12 example

WHAT YOU SEE

Juror holding 'MELD 12'

WHAT IT ENCODES

Worked example: a MELD of 12 is below the 15 threshold, so the disposition is to continue to wait, optimize medically, and re-score periodically rather than transplant.

BOARD TRAP

WRONG answer: prioritizing the MELD-12 patient over someone with a higher MELD or an HCC exception. Discriminator: 12 is below the benefit threshold — they wait; a MELD-12 with no exception does not outrank higher-MELD or exception-point candidates.

12. Encephalopathy crossed (MELD)

WHAT YOU SEE

'CHILD-PUGH' with encephalopathy crossed under MELD judge

WHAT IT ENCODES

This crossed-out item reinforces that MELD deliberately EXCLUDES the subjective encephalopathy and ascites that Child-Pugh relies on, keeping allocation objective and reproducible across centers.

BOARD TRAP

WRONG answer: expecting a patient's encephalopathy/ascites to raise their MELD score. Discriminator: those are Child-Pugh-only; to capture sickness MELD instead uses creatinine, bilirubin, INR and sodium — clinical decompensation may not move the MELD number.

13. Right judge: Factor V

WHAT YOU SEE

'RIGHT JUDGE: Factor V judge', 'BEST ACUTE MARKER'

WHAT IT ENCODES

Factor V is the BEST marker of synthetic function in ACUTE liver failure: it is liver-synthesized, has a short half-life (~12-36 h) so it tracks rapid changes, and is NOT vitamin-K-dependent so it isn't confounded by vitamin K status. In fulminant failure a very low Factor V predicts poor prognosis (a component of the King's College / Clichy criteria).

BOARD TRAP

WRONG answer: using albumin or even raw PT/INR as the best acute synthetic marker. Discriminator: albumin is too slow (~20-day half-life) and PT/INR is confounded by vitamin K/transfusion — Factor V is the cleanest real-time readout of acute hepatic synthesis.

14. Factor V chained

WHAT YOU SEE

Factor V vial chained, 'PT-INR-independent'

WHAT IT ENCODES

Factor V reflects TRUE liver synthesis and is independent of vitamin K — unlike the vitamin-K-dependent factors (II, VII, IX, X) that prolong PT/INR. So if a prolonged PT corrects after vitamin K, the problem was deficiency/malabsorption; if Factor V is low, the liver itself is failing to synthesize.

BOARD TRAP

WRONG answer: attributing a low Factor V to vitamin K deficiency. Discriminator: Factor V does NOT depend on vitamin K and will NOT correct with it — a low Factor V means genuine hepatic synthetic failure, the key distinction from cholestatic vitamin-K malabsorption.

15. PT/INR useless acute

WHAT YOU SEE

'PT/INR USELESS ACUTE' crossed

WHAT IT ENCODES

In ACUTE liver failure, PT/INR is confounded — vitamin K administration, fresh frozen plasma/transfusion, and concurrent coagulopathy all distort it — so it is unreliable for tracking synthetic function moment-to-moment. Factor V is preferred in that setting.

BOARD TRAP

WRONG answer: trusting a 'corrected' INR after FFP/vitamin K as evidence the liver is recovering. Discriminator: FFP and vitamin K artificially normalize INR without improving hepatic synthesis — follow Factor V instead to judge true recovery in acute failure.

16. Albumin too slow

WHAT YOU SEE

'ALBUMIN 18-20 days TOO SLOW'

WHAT IT ENCODES

Albumin has a long serum half-life (~18-21 days), so it changes too slowly to reflect ACUTE liver failure — it is a useful marker of CHRONIC synthetic function (and a Child-Pugh component) but lags behind acute events by weeks.

BOARD TRAP

WRONG answer: using albumin to assess or trend acute/fulminant liver failure. Discriminator: albumin's ~20-day half-life makes it useless acutely (and it is also lowered by nephrotic syndrome, malnutrition, inflammation) — use Factor V or PT/INR-context for acute synthetic assessment, reserve albumin for chronic.

17. Child-Pugh juror cards

WHAT YOU SEE

Jurors holding 'CHILD-PUGH 7' cards

WHAT IT ENCODES

The jurors holding 'Child-Pugh 7' represent class B (≥ 7), the chronic-disease threshold that prompts transplant EVALUATION/referral. It anchors the chronic side of the court (Child-Pugh for chronic decompensated cirrhosis) against the acute side (Factor V for acute liver failure).

BOARD TRAP

WRONG answer: confusing the indications — applying Child-Pugh/MELD logic to acute liver failure or Factor V/King's College logic to chronic cirrhosis. Discriminator: chronic decompensated cirrhosis is staged/referred by Child-Pugh and listed by MELD-Na; ACUTE liver failure is assessed by Factor V and King's College criteria — and recall transplant indications (decompensated cirrhosis, acute liver failure, HCC within Milan criteria: single tumor ≤ 5 cm or up to 3 tumors each ≤ 3 cm, no vascular invasion) versus contraindications (active extrahepatic malignancy, uncontrolled sepsis, severe cardiopulmonary disease, active substance use).
